



KENYA REGIONAL CROP YIELD PREDICTION



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- Agriculture drives Kenya's GDP, food security, and employment.
- Yield variability caused by climate, inputs, and regional disparities.
- Current planning relies on historical trends.



Problem Context



Problem Statement & Objectives

- Predict regional crop yields using climate & input data.
- Identify climate-sensitive regions.
- Evaluate impact of pesticides & harvested area.
- Provide early warning for food shortages.



Data Sources

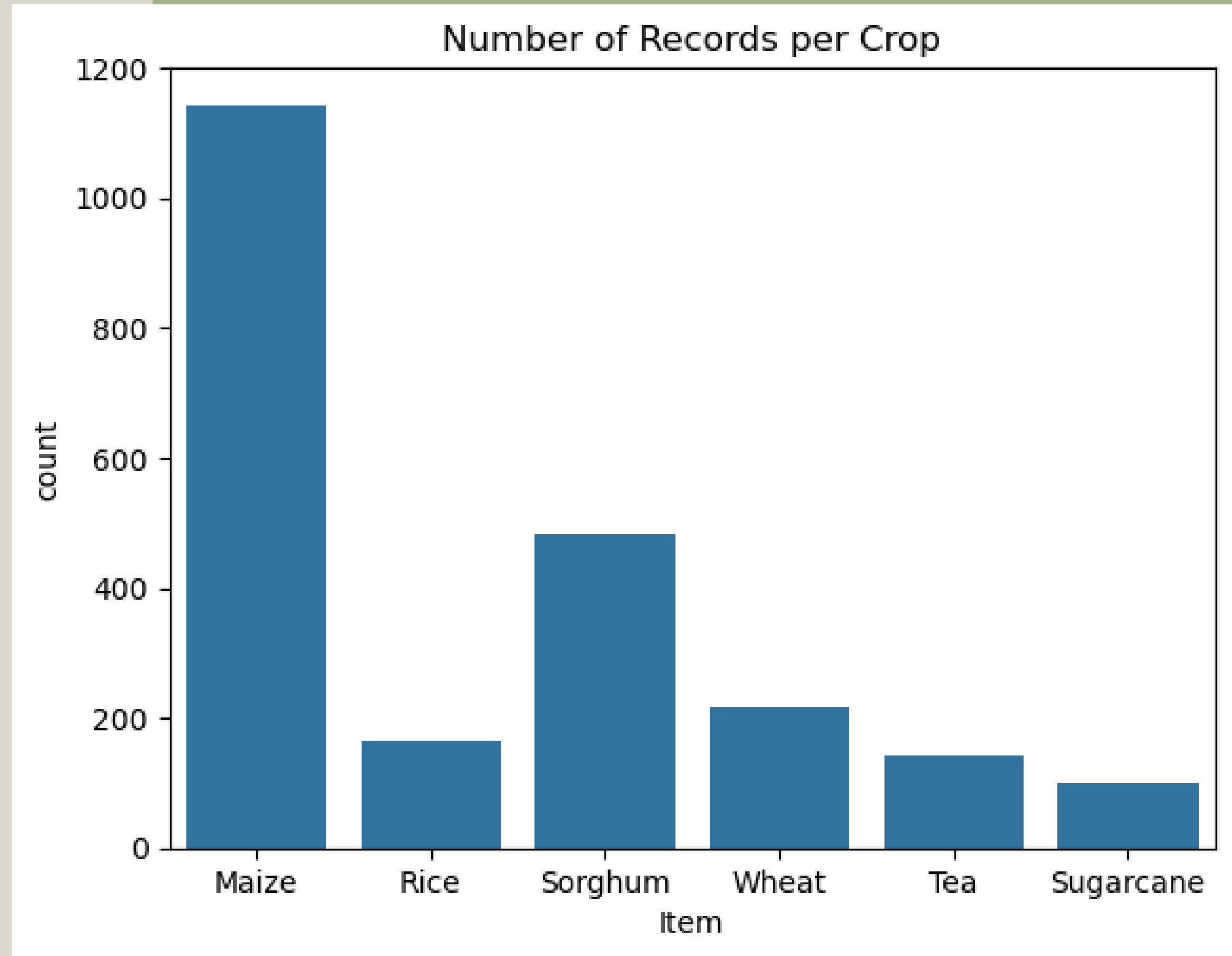
- Kenya Agricultural Production Dataset (Kaggle)
- Regional Crop Production Dataset (HarvestStat Africa)
- Climate Data - Rainfall & Temperature (OpenAfrica)
- Pesticide Usage Dataset (KAPSARC Data Portal)



Data Integration & Features

- Merged datasets by year, region, crop.
- Created seasonal climate features
- Imputed missing climate & pesticide values.
- Final features include:
 - Rainfall
 - Temperature
 - Area harvested
 - Pesticide usage
 - Region, crop, season

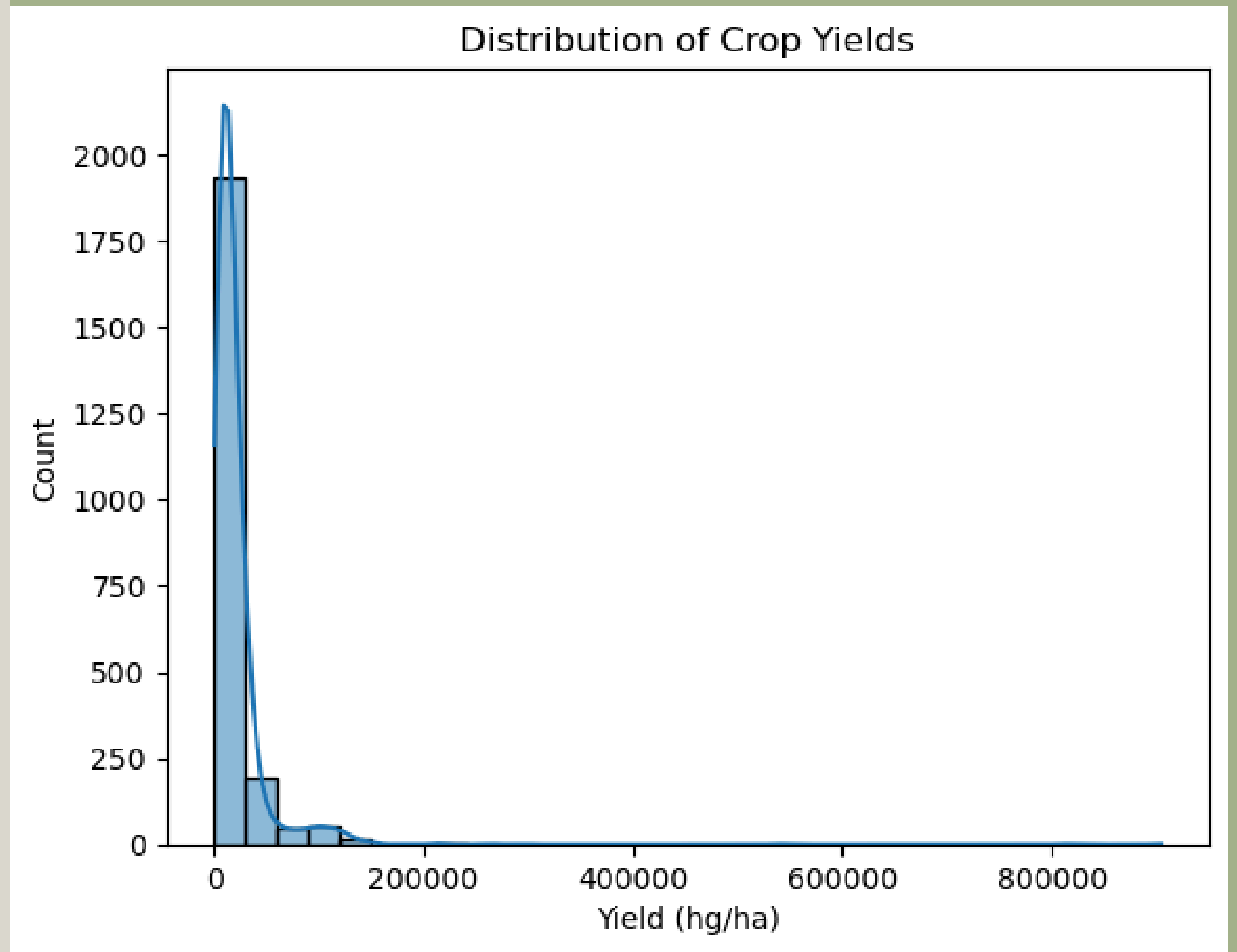
Crop Distribution



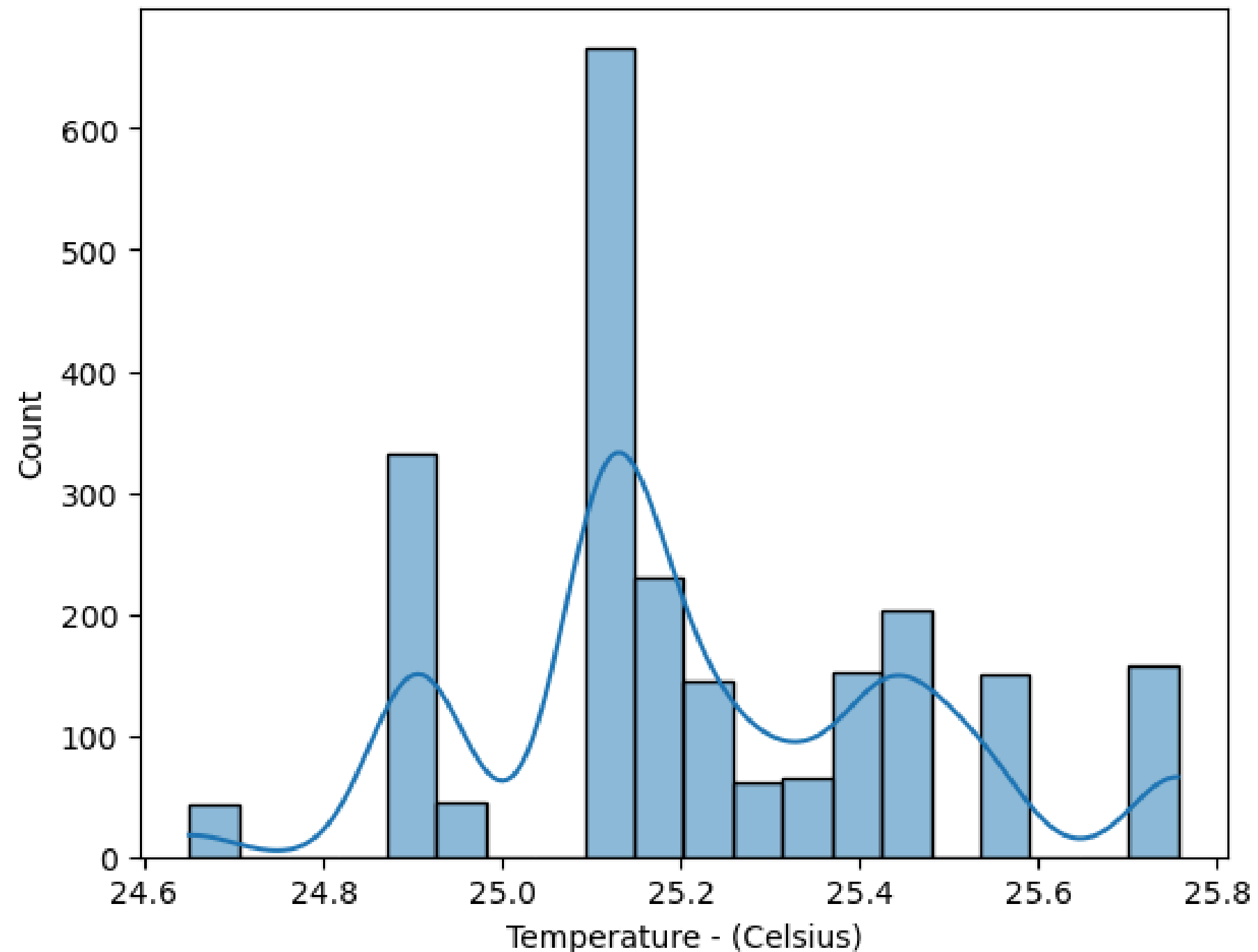
- Maize dominates dataset.
- Potential model bias toward major staple crops

Yield Distribution

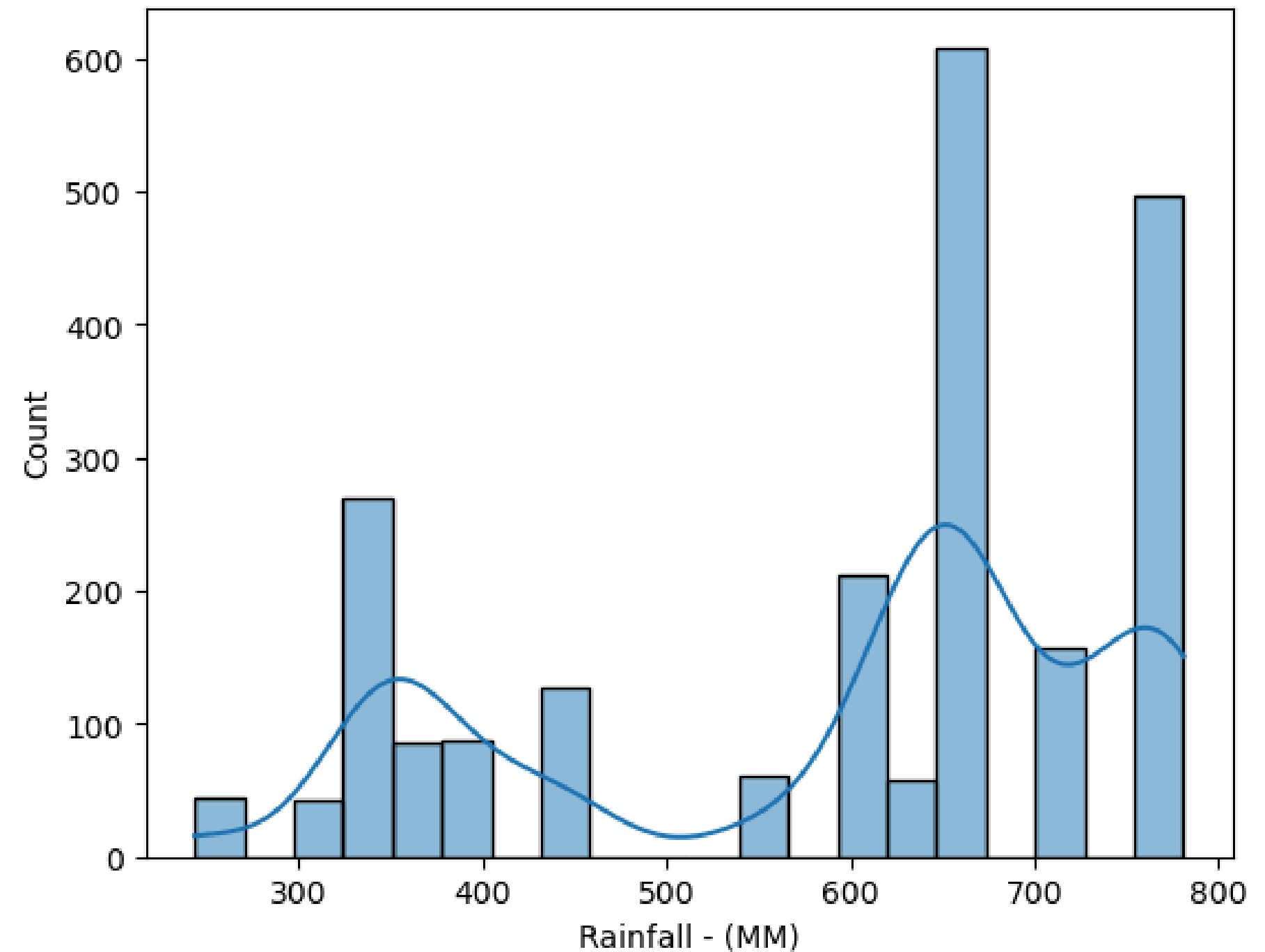
- Right-skewed distribution.
- Most yields fall below ~180,000 hg/ha.
- Presence of extreme values this justified using robust models.



Temperature Distribution (°C)

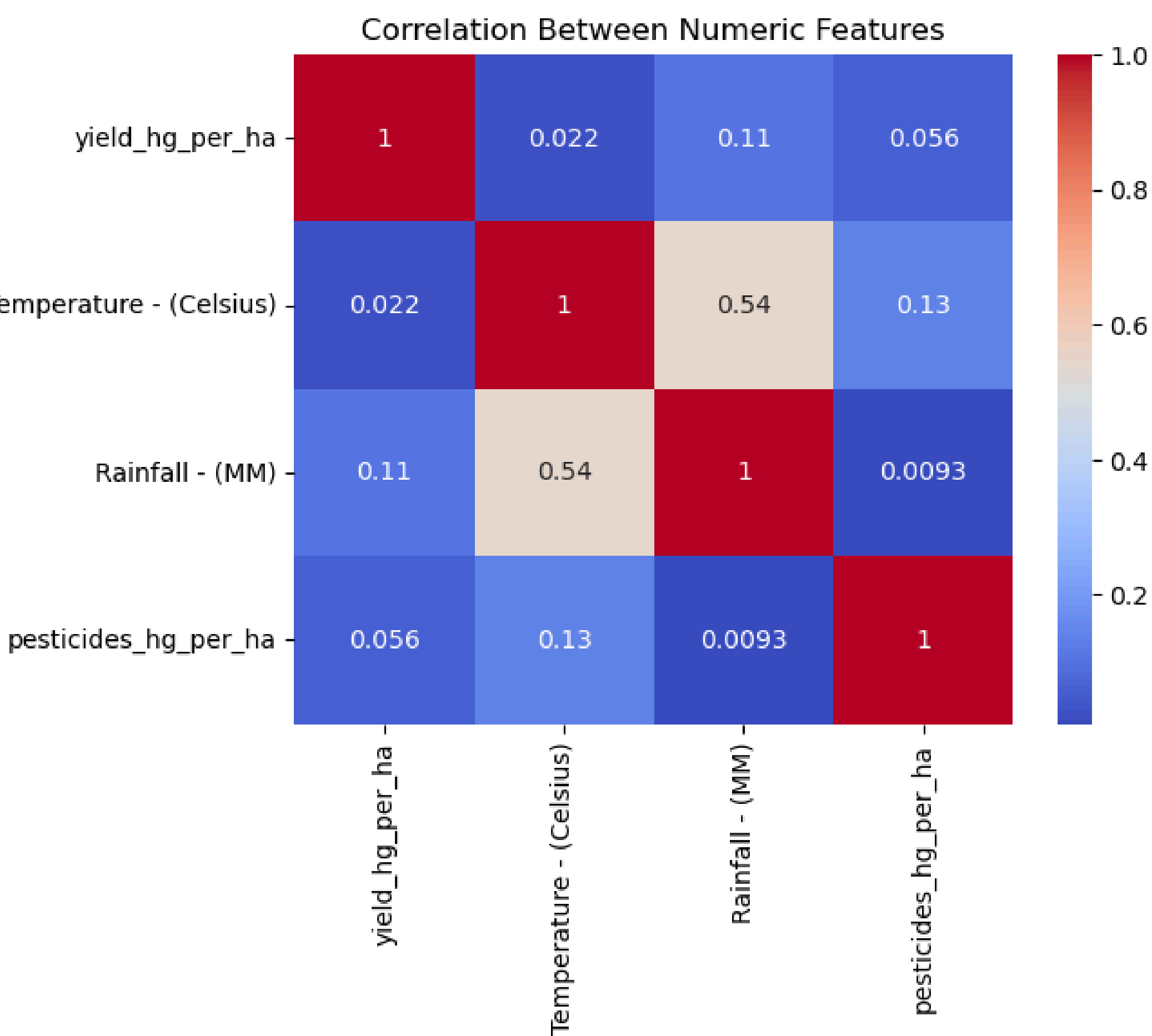


Rainfall Distribution (mm)



Climate Distribution

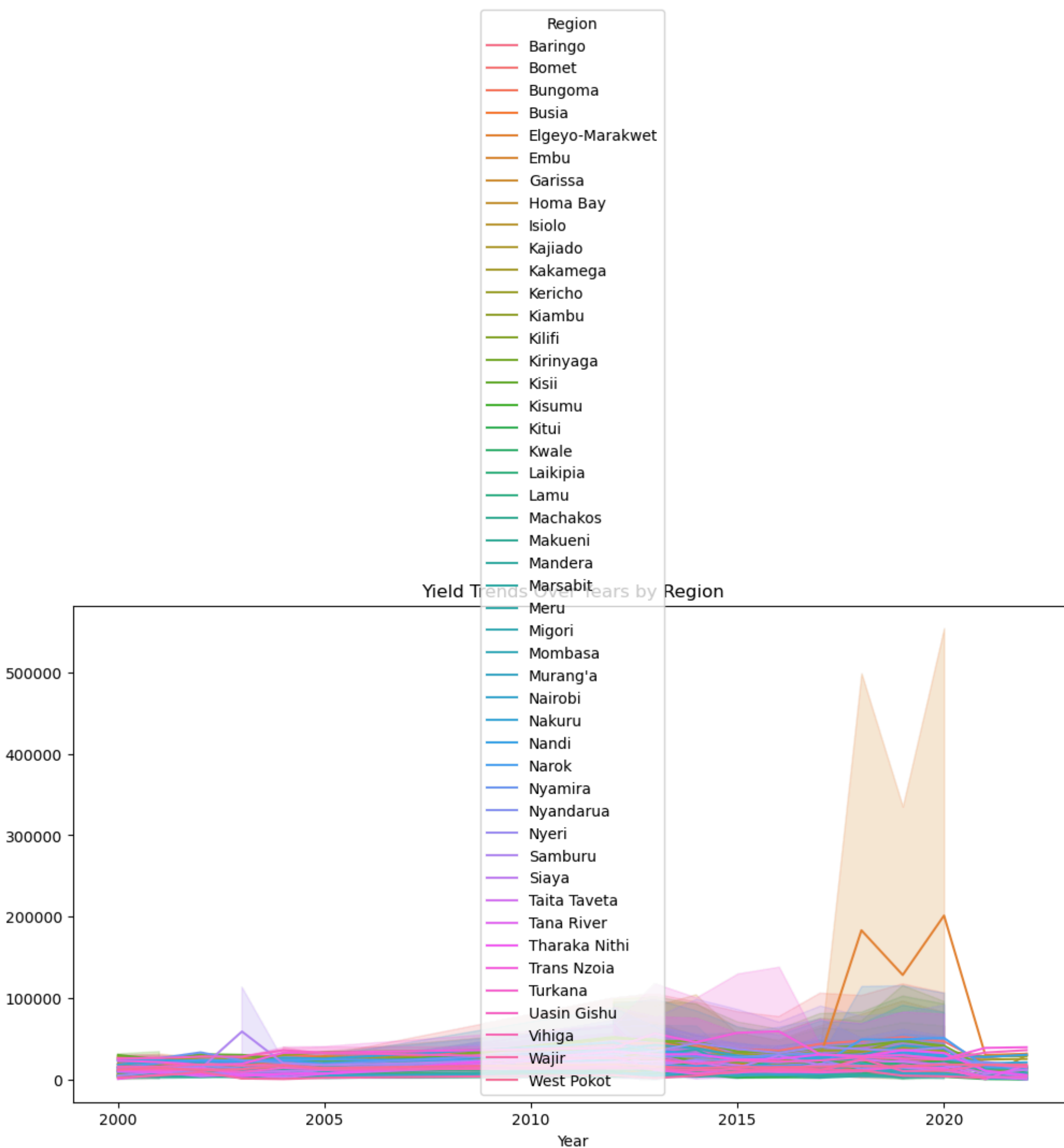
- Temperature shows low variability.
- Rainfall varies significantly.
- Rainfall likely stronger predictor.



Correlation Heatmap

- Weak direct correlations, complex relationships.
- Justifies nonlinear models like Random Forest.

Yield Trends by Region



- Regional disparities in yield patterns.
- Some regions show volatility likely climate sensitivity.



Modeling Approach

- Models tested:
 - Linear Regression
 - Random Forest
 - XGBoost
- Purpose:
 - Compare linear vs nonlinear performance.



Model Performance Comparison

MODEL	R^2	RMSE	MAE
Linear Regression	0.372	1.649	0.915
Random Forest	0.716	1.109	0.609
XGBoost	0.700	1.140	0.656

- Random Forest best generalization.
- Linear regression underfits.



Why Random Forest Won

- Captures nonlinear relationships.
- Handles region × crop interactions
- Robust to noisy & imputed data.
- Lower overfitting risk than XGBoost.



Key Insights

- Rainfall is the strongest climate driver.
- Some regions are highly climate-sensitive.
- Pesticide use shows moderate influence.
- Yield prediction enables proactive planning.



Business Impact

Model enables:

- Early food shortage warnings.
- Climate risk planning.
- Region-specific agricultural policy.
- Smarter resource allocation.



Future Work

- Add satellite & soil data.
- Incorporate irrigation data.
- Expand to county-level predictions.



Flask app

Kenya Crop Yield Predictor

This tool predicts crop yield (tons per hectare) across Kenyan regions using a machine learning model trained on historical agricultural data. The model analyzes environmental and farming factors such as rainfall, temperature, pesticide usage, crop type, season, and year to estimate expected yield.

Region:

Crop:

Year:

Season:

Area (ha):

Production (tons):

Temperature - (Celsius):

Rainfall (mm):

Pesticides_hg_per_ha:

- A deployed Flask-based web application for predicting crop yield in Kenyan regions.
- Provides a user-friendly interface for non-technical users to access machine learning predictions.
- Connects a trained Random Forest model to real-world decision-making.

Conclusion



Although the model exhibits potential as a yield risk indicator, proxy input variables, climate aggregation, and data granularity limit its predictive ability. As a result, rather than being a precise forecasting system, it should be understood as a decision-support tool.

Thank You.



TEAM MEMBERS:

- SHEILLA MACHARIA
- MARGARET MONDIA
- MESHACK MBOYA
- BENARD KARIUKI
- ISAAC NDUNGU
- BRAVIAN ASHONO